M&AE 4070/5070 - Dynamics of Flight Vehicles

Case Study 2 Solutions

Descent and approach of a commercial airplane

This case study explores how an aircraft transitions from cruise to descent, followed by approach as it gets closer to the airport. We treat each of these 3 phases (cruise, descent, approach) assuming pure longitudinal flight at equilibrium (cruise) or pseudo-equilibrium (descent/approach). For all calculations, we will consider a generic commercial transport aircraft, whose characteristics will be provided below.

The objective is to understand how the transition from one phase of flight to another can be done using the available "pilot" and "configuration" controls. In this case, the "pilot" controls are pitch control δ_m and throttle δ_x, and the "configuration" controls are the air brakes (also called speed brakes), δ_B, spoilers, δ_{SP}, slats, δ_S, and flaps, δ_F. The last two are usually coupled, and controlled together according to an aircraft-specific law, which we'll call δ_{SF}. **We want to evaluate how each control, in particular the air/speed brakes and spoilers, impacts steady-state flight, and understand how the controls can be used to successfully accomplish each phase of flight**

The physical difference between air brakes and spoilers is often fuzzy, especially since both can refer to the exact same moving surfaces in many aircrafts. In MAE 4070/5070, we will consider that

- **Air (or speed) brakes** are the moving surfaces located on the upper surface of the wings (often more inward than spoilers), deploying **up**. They are used **in flight** with small deflection angles, to increase drag while keeping lift virtually unchanged.
- **Spoilers**, on the other hand, are used during landing, and therefore, much closer to the ground, typically just after wheel touch-down. Deployed with a much larger deflection angle, their main effect is to destroy lift while increasing drag, thereby changing the lift-to-drag ratio.

Models for how each of those change the lift, drag, and pitching moment coefficients will be provided in the "Available data" section.

The case study is organized in different parts, one for each stage of flight:

1. You will first characterize cruise conditions.
2. Then you will explore the options available to the pilot to safely descend from cruise altitude to around 1000m.
3. Once at 1000m, you will characterize how the deployment of slats, flaps, and landing gear allow the aircraft to intercept the ILS beam and place itself on its final trajectory before landing.
4. The last part looks at this final trajectory along the ILS beam before landing.

Available data. We consider a typical medium-haul transport aircraft, and describe the aircraft's geometry, engines and force and moment coefficients using the following models:

- **Geometric and mass data.**

Parameter	Symbol and value
Reference surface	$S = 260$ m^2
Reference length	$\ell = 6.6$ m
Effective aspect ratio	$\mathbf{AR} = 5.78$
Mass at the end of cruise	$m_f = 1.1 \times 10^5$ kg $= 110$ tons
Maximal mass at take-off	$m_i = 1.6 \times 10^5$ kg $= 160$ tons

- **Engines data.**

  This aircraft is equipped with two jet engines located under each wing.

Parameter	Symbol and value
Reference thrust (sea level) per engine	$F_0 = 1.2 \times 10^5$ N
Altitude-dependent total thrust from both engines	$F = 2\frac{\rho}{\rho_0} F_0 \delta_x$

- **Lift coefficient C_z.** The lift coefficient is written:

$$C_z = C_{z_\alpha} \bar{\alpha} + C_{z_{\delta_m}} \delta_m + \Delta C_z \tag{1}$$

  – The term $C_{z_\alpha} \bar{\alpha}$ gives the C_z that one would obtain in a wind tunnel for example. That is often the C_z provided in the drag polar curves. We need to add to that the contribution from the elevator (described by δ_m), whose position is adjusted to ensure that $C_m = 0$ during equilibrium/pseudo-equilibrium flights.

  – A (negative) ΔC_z term is added to account for loss of lift due to spoiler deployment.

- **Drag coefficient C_x.** The drag coefficient is written as:

$$C_x = C_{x_0} + \Delta C_{x_0} + k C_z^2 \tag{2}$$

  – The increase in drag ΔC_{x_0} comes from the deployment of landing gear, air brakes, spoilers, slats, and flaps. The corresponding values are summarized below.

  – The k coefficient is linked to the aircraft aspect ratio and comes for the most part from induced drag. Here, we model it simply as:

$$k = \frac{1}{\pi \mathbf{AR}} \tag{3}$$

- **Pitching moment coefficient C_m.** We model the pitching moment coefficient as a linear function of the elevator position δ_m and $\bar{\alpha}$, the difference between the angle of attack and the angle of attack at zero lift:

$$\bar{\alpha} = \alpha - \alpha_0 \,. \tag{4}$$

  The moment coefficient can therefore be written as:

$$C_m = C_{m_0} + \Delta C_{m_0} + C_{m_\alpha} \bar{\alpha} + C_{m_{\delta_m}} \delta_m \tag{5}$$

  – At steady or pseudo-steady state, C_m must be equal to zero.

  – $C_m = C_{m_0}$ when the angle of attack is equal to α_0 and $\delta_m = 0$ in the reference aircraft configuration.

  – The increase in pitching moment ΔC_{m_0} is due to the deployment of air brakes, spoilers, slats and flaps.

The value of all constant coefficients and partial derivatives in the models above, along with the effect of slats and flaps on max angle of attack and angle of attack at zero lift are summarized below.

Configuration	C_{z_α}	$C_{z_{\delta_m}}$	C_{x_0}	k	C_{m_0}	C_{m_α}	$C_{m_{\delta_m}}$
Value	5	0.44	0.0175	0.055	-0.1	-1	-1.46

Table 1: Aircraft coefficients

Configuration	$\alpha_{\max}$	α_0	$\bar{\alpha}_{\max}$
Nothing deployed	0.244 rad = 14°	-0.035 rad = -2°	0.28 rad = 16°
With slats and flaps	0.314 rad = 18°	-0.262 rad = -15°	0.576 rad = 33°

Table 2: Angle of attack

Configuration	Notation	ΔC_{x_0}	ΔC_z	ΔC_{m_0}
Landing gear	LG	0.0175	0	0
Air brakes	B	0.036	0	0
Spoilers - cruise	SP	0.036	-0.05	0.03
Spoilers - landing	SP_L	0.06	-0.15	0.09
Slats and flaps	SF	0.06	0	-0.3

Table 3: Incremental change in lift, drag and moment coefficients due to change in configurations. Slats and flaps are deployed at 25°. Obviously, deploying slats and flaps does change C_z, but this is accounted for through the change in $\alpha_{\max}$. For spoilers, a $\sim 40^o$ deflection is assumed.

1 Cruise conditions

1. The aircraft is cruising at an altitude of 10 km and a Mach number of 0.8. Characterize the steady-state corresponding to cruising conditions, that is, calculate $C_{z_{cr}}$, $C_{x_{cr}}$, throttle $\delta_{x_{cr}}$, and position of the elevator $\delta_{m_{cr}}$. Comment on the value of $\delta_{x_{cr}}$.

Answer.

- The Mach number is defined as the ratio between the velocity and the speed of sound a:

$$a = \sqrt{\gamma RT} = \sqrt{\gamma \frac{p}{\rho}}, \text{ using the ideal gas law to convert one form into the other.} \tag{6}$$

$$V_{cr}^2 = a^2 M_{cr}^2 = \gamma \frac{p}{\rho} M_{cr}^2 \tag{7}$$

- The pressure and density at cruise altitude is obtained from the integrated hydrostatic equation, assuming linear decrease of the temperature with the altitude:

$$p_{cr} = p_{\text{SSL}} \left(1 + \frac{ah}{T_{\text{SSL}}}\right)^{-\frac{g}{aR}} = 101,325 \left(1 + \frac{-0.0065 \times 10,000}{288.15}\right)^{\frac{9.81}{0.0065 \times 287}} = 2.6418 \times 10^4 \text{ Pa} \tag{8}$$

$$\rho_{cr} = \rho_{SSL} \left(1 + \frac{ah}{T_{\text{SSL}}}\right)^{-\frac{g}{aR}-1} = 0.412 \text{ kg/m}^3 \tag{9}$$

- The cruise **lift coefficient** $C_{z_{cr}}$ is determined from equation $\boxed{\mathbf{S}}$:

$$C_{z_{cr}} = \frac{m_f g}{\frac{\gamma}{2} \rho S V_{cr}^2} = \frac{mg}{\frac{\gamma}{2} p_{cr} S M_{cr}^2} = \frac{1.1 \times 10^5 \times 9.81}{\frac{1.4}{2} \frac{101,325}{4} 260 \times 0.8^2} = 0.35 \tag{10}$$

  where the mass is taken at the end of the cruise portion of the flight (*i.e.* $m = m_f$). This is the correct order of magnitude for a cruising lift coefficient.

- The **drag coefficient** is evaluated from the drag polar:

$$C_{x_{cr}} = C_{x_0} + k C_{z_{cr}} = 0.0175 + 0.055 \times 0.35^2 = 0.024 \tag{11}$$

  The drag force, which we will need to evaluate δ_x, is then:

$$D_{cr} = \frac{\gamma}{2} p_{cr} S M_{cr}^2 C_x = \frac{m_f g}{C_{z_{cr}}} C_{x_{cr}} = 7.46 \times 10^4 N \tag{12}$$

- **Thrust** can be expressed as a function of throttle δ_x:

$$F_{cr} = 2 \frac{\rho}{\rho_0} F_0 \delta_x = 2 \frac{0.412}{1.225} 1.2 \times 10^5 \delta_x = 8.08 \times 10^4 \delta_x N \tag{13}$$

  Equation $\boxed{\mathbf{P}}$ for cruise conditions shows that drag must be equal to thrust. We can therefore evaluate δ_x:

$$F_{cr} = D_{cr} \tag{14}$$

$$8.08 \times 10^4 \delta_x = 7.46 \times 10^4 \tag{15}$$

$$\delta_x = \frac{7.46}{8.08} = 0.923 = 92.3\% \text{ of max throttle} \tag{16}$$

  The engines are dimensioned for cruise conditions, since they are close to their max at cruising conditions.

- **The position of the elevators**, δ_m, can be determined from the coupled set of equations $\boxed{\mathbf{S}}$ and $\boxed{\mathbf{M}}$, using the proper model for $C_{z_{cr}}$:

$$C_{z_{cr}} = C_{z_\alpha}\bar{\alpha}_{cr} + C_{z_{\delta_m}}\delta_{m_{cr}} \tag{17}$$
$$0 = C_{m_0} + C_{m_\alpha}\bar{\alpha}_{cr} + C_{m_{\delta_m}}\delta_{m_{cr}} \tag{18}$$

yielding

$$\delta_{m_{cr}} = \frac{C_{m_0} + \frac{C_{m_\alpha}C_{z_{cr}}}{C_{z_\alpha}}}{\frac{C_{z_{\delta_m}}C_{m_\alpha}}{C_{z_\alpha}} - C_{m_{\delta_m}}} = \frac{-0.1 + \frac{-1\times 0.35}{5}}{\frac{-1\times 0.44}{5} + 1.46} = -0.124 \tag{19}$$

Assuming the non-dimensional coefficients are scaled with $\bar{\alpha}$ and δ_m expressed as angles, the deflection of the elevator during cruise is therefore -0.124 rad or -7.1°. The angle attack can also be calculated:

$$\alpha_{cr} = \bar{\alpha}_{cr} + \alpha_0 = \frac{1}{C_{z_\alpha}}\left(C_{z_{cr}} - C_{z_{\delta_m}}\delta_{m_{cr}}\right) + \alpha_0 = \frac{1}{5}\left(0.35 - 0.44\times 0.124\right) - 0.035 = 0.081 \text{ rad} = 4.64^\text{o} \tag{20}$$

2 From cruise to 1000 m

The aircraft is getting close to its destination, and the pilot wants to initiate the descent. At this altitude and Mach number, the aircraft is close to its Maximum Operating velocity, $V_{MO} \sim 350$ knots (*calibrated* airspeed) and Mach number $M_{MO} \sim 0.87$. The limit in velocity is linked to the structural integrity of the aircraft, while the limit in Mach number is linked to buffeting. The pilot therefore wants to choose a way to descend that **will not increase the aircraft velocity**. His choices in terms of controls are the following:

- reduce throttle, decreasing δ_x (fully reduced throttle would correspond to a value $\delta_x = 0.06 = 6\%$)
- act on the stick, that is, change the elevator position and δ_m.
- deploy the air brakes, affecting δ_B,
- deploy the spoilers by increasing δ_{SP}.

To make an informed choice about which strategy to recommend, we need to analyze the new steady state obtained after having changed one of the controls. It is easier if only one control is changed at a time, independently from the others.

2. **Changing δ_m to initiate the descent.** Qualitatively explain what would happen if you change δ_m, that is, if you modify the stick position.

   Answer.

   From the cruise steady-state, the pilot can initiate its descent by pushing on the stick: that positive change in δ_m will increase the velocity, which will increase drag (since the aircraft is flying at cruise velocity, we are in the first regime) and create a negative Thrust-Drag balance. The increase in velocity comes from $\boxed{\mathbf{S}}$ and the decrease in angle of attack. This may be problematic, as we are already very close from the max operating Mach number.

3. **Descending by reducing throttle δ_x.** Calculate the descent slope and vertical velocity of the aircraft if the pilot decreases throttle to 6%.

Answer.

Fully reduced throttle corresponds to $\delta_x = 0.06 = 6\%$, in which case thrust will be:

$$F_{de} = 2\frac{\rho_{cr}}{\rho_0} F_0 \delta_x = 4.858 \times 10^3 N \tag{21}$$

Drag has been calculated already (since the stick has not moved, the drag has not changed either). The propulsion equation therefore gives us:

$$F_{de} - D_{cr} = mg\gamma_{de} \tag{22}$$

$$\gamma_{de} = -0.0646 \text{ rad} = -3.70^{\text{o}} \tag{23}$$

The vertical velocity is (with a speed of sound of $\sqrt{\gamma p_{cr}/\rho_{cr}} = 299$ m/s at 10km and a Mach number of 0.8):

$$w_{de} = V_{cr}\gamma_{de} = -299 \times 0.8 \times 0.0646 = -15.48 \text{ m/s} \tag{24}$$

This is a realistic descent rate, and this approach is the one typically used in practice. Reducing throttle yields an efficient descent, cost-effective and easy to pilot.

4. **Descending by deploying airbrakes.** Calculate the descent slope and vertical velocity of the aircraft if the pilot deploys the air brakes.

Answer.

The propulsion equation including air brakes is as follows:

$$F_{cr} - \frac{\gamma}{2} p_{cr} S M_{cr}^2 \left(C_{x_{cr}} + \Delta C_{x_{0,B}} \right) = m_f g \gamma_B \tag{25}$$

Since thrust is equal to drag at cruise conditions:

$$F_{cr} = \frac{\gamma}{2} p_{cr} S M_{cr}^2 C_{x_{cr}} \,, \tag{26}$$

the slope will be

$$\gamma_B = -\frac{\frac{\gamma}{2} p_{cr} S M_{cr}^2 \Delta C_{x_{0,B}}}{m_f g} = -0.1027 \text{ rad} = -5.88^{\text{o}} \tag{27}$$

The vertical velocity is:

$$w_B = V_{cr} \gamma_B = -299 \times 0.8 \times 0.1027 = -24.59 \text{ m/s} \tag{28}$$

With this rate of descent, we would need less than 7 minutes to reach ground level (and the passengers would heavily complain about their ears).

5. **Descending by deploying spoilers.** Calculate the descent slope and vertical velocity of the aircraft if the pilot fully deploys the spoilers, leaving the stick unchanged from the cruise configuration. Comment on the value of C_z during this descent compared to its value during cruise: does that correspond to what you expected? Why?

Answer.

The pilot deploys the spoilers, and maintains δ_m at the cruise value $\delta_{m_{cr}}$. To find the new Mach number, we need the new $C_{z_{SP}}$, and solve the sustentation equation.

The change in C_{m_0} is $\Delta C_{m_{0,SP}} = 0.036$, which creates a change of angle of attack visible through the moment equation $\boxed{\mathbf{M}}$:

$$C_{m_0} + \Delta C_{m_{0,SP}} + C_{m_\alpha}\bar{\alpha}_{SP} + C_{m_{\delta_m}}\delta_{m_{cr}} = 0 \tag{29}$$

Therefore, the new lift coefficient $C_{z_{SP}}$ depends on this new angle of attack $\bar{\alpha}_{SP}$, and of the position of the pitch control during cruise, which was evaluated earlier:

$$C_{z_{SP}} = C_{z_\alpha}\bar{\alpha}_{SP} + C_{z_{\delta_m}}\delta_{m_{cr}} + \Delta C_{z_{SP}}\,. \tag{30}$$

$\boxed{\mathbf{M}}$ allows us to extract the new value for the angle of attack $\bar{\alpha}_{SP}$:

$$\bar{\alpha}_{SP} = -\frac{1}{C_{m_\alpha}}\left(C_{m_0} + \Delta C_{m_{0,SP}} + C_{m_{\delta_m}}\delta_{m_{cr}}\right) \tag{31}$$

$$= -0.1 + 0.03 - 1.46 \times (-0.124) \tag{32}$$

$$= 0.111 \text{ rad} = 6.36^{\text{o}} \tag{33}$$

The new lift coefficient can therefore be calculated as:

$$C_{z_{SP}} = C_{z_\alpha}\bar{\alpha}_{SP} + C_{z_{\delta_m}}\delta_{m_{cr}} + \Delta C_{z_{SP}} \tag{34}$$

$$= 5 \times 0.111 + 0.44 \times (-0.124) - 0.05 \tag{35}$$

$$= 0.45 \tag{36}$$

C_z has been increased by 0.1 compared to cruise, even though the direct effect of spoilers is to decrease this coefficient by 0.05. The indirect effect of the moment coefficient is to increase the angle of attack (nose up), which also increases C_z . The velocity for the descent therefore decreases, as shown by $\boxed{\mathbf{S}}$:

$$M_{SP} = \sqrt{\frac{m_f g}{0.7 p_{cr} S C_{z_{SP}}}} = 0.706 \tag{37}$$

This decrease in velocity provides an additional margin which could allow to push on the stick to increase the slope. Finally, one last thing we can calculate is the value of the slope γ obtained when deploying the spoilers, and compare that to the approaches above.

Let's look at Eq. $\boxed{\mathbf{P}}$. Since throttle (or δ_x) has not been touched, thrust has not changed. However, drag needs to be recalculated with the new $C_{z_{SP}}$ and the extra drag coming from the spoilers.

$$C_{x_{SP}} = C_{x_0} + \Delta C_{x_{SP}} + k C_{z_{SP}}^2 = 0.0175 + 0.036 + 0.055 \times 0.45^2 = 0.0646 \tag{38}$$

$$D_{SP} = 0.7 p S M_{SP}^2 C_{x_{SP}} = 1.55 \times 10^5 N \tag{39}$$

The slope is then evaluated from $\boxed{\mathbf{P}}$:

$$\gamma_{SP} = \frac{F_{cr} - D_{SP}}{m_f g} = -0.074 \text{ rad} = -4.26^{\text{o}} \tag{40}$$

Spoilers end up yielding a slope $\sim 50\%$ lower than that of the airbrakes, but the descent also occurs at a slower velocity (Mach of 0.71 instead of 0.8). The corresponding vertical velocity is:

$$w_{SP} = V_{SP}\gamma_{SP} = -299 \times 0.706 \times 0.074 = -15.62 \text{ m/s} \tag{41}$$

For a fair comparison, one should "push on the stick" to recover the same Mach, and re-evaluate the slope.

6. **Summary.** Using the data that you calculated and some practical considerations, provide a recommendation to the pilot.

Answer.

The pilot should use the throttle only, as it is easier to control, is cost effective, and when engines are at minimum (but not zero, you don't want to deal with engine restart in the air), yields a very reasonable descent slope.

3 Deploying slats, flaps, and landing gear

After the descent phase, the pilot stabilizes the aircraft trajectory at a constant altitude of 1000 m, and adjusts its velocity to an appropriate approach velocity $V_{ap} = 99.5$ m/s = 193 kt, a value 30% higher than stall velocity (without flaps and slats deployed yet, the available data yield a $C_{z_{\max}} = 1.28$). He then deploys landing gear, slats and flaps to intercept the ILS beam: we want to quantify the effect of each of those on the flight parameters.

7. What is the density at this new altitude?

Answer.

Using the same equation as in the very first part:

$$\rho_{ap} = \rho_{SSL}\left(1 + \frac{ah_{ap}}{T_{\text{SSL}}}\right)^{-\frac{g}{aR}-1} = 1.11 \text{ kg/m}^3 \tag{42}$$

8. Calculate throttle δ_x, position of the elevator δ_m and angle of attack $\bar{\alpha}$ at the 1000 m steady-state, before deploying anything.

Answer.

- The stall velocity is obtained from $\boxed{\mathbf{S}}$ evaluated with $C_{z_{\max}}$, and the approach velocity by adding 30% to it:

$$V_S^2 = \frac{mg}{\frac{1}{2}\rho S C_{z_{\max}}} = 5847 \tag{43}$$

$$V_S = 76.5 m/s = 148.5\, kt \tag{44}$$

$$V_{ap} = 1.3 V_S = 99.5 m/s = 193\, kt \tag{45}$$

- **Initial drag:** To get the drag force, we need C_x, and therefore C_z. Taking the ratio between $\boxed{\mathbf{S}}$ evaluated at stall and $\boxed{\mathbf{S}}$ evaluated at the current velocity, we get:

$$\left(\frac{V}{V_S}\right)^2 = \frac{C_{z_{\max}}}{C_z} = 1.3^2 = 1.69 \tag{46}$$

$$Cz = \frac{C_{z_{\max}}}{1.69} = \frac{1.28}{1.69} = 0.757 \tag{47}$$

and

$$C_x = C_{x_0} + kC_z^2 = 0.0175 + 0.055 \times 0.757^2 = 0.049 \tag{48}$$

$$D = \frac{1}{2}\rho S V^2 C_x = \frac{1.11}{2} 260 \times 99.5^2 \times 0.049 = 7.44 \times 10^4 N \tag{49}$$

Thrust can be expressed as

$$F = 2\frac{\rho}{\rho_0} F_0 \delta_x = 2.175 \times 10^5 \delta_x N \tag{50}$$

hence the position of the throttle control:

$$\delta_x = \frac{0.744}{2.175} = 0.322 = 32\% \text{ of max thrust} \tag{51}$$

- **angle of attack and δ_m control:** as above, a system of equations needs to be solved:

$$C_m = C_{m_0} + C_{m_\alpha}\bar{\alpha} + C_{m_{\delta_m}}\delta_m = 0 \tag{52}$$

$$C_z = C_{z_\alpha}\bar{\alpha} + C_{z_{\delta_m}}\delta_m\,, \tag{53}$$

or

$$-\bar{\alpha} - 1.46\delta_m = 0.1 \tag{54}$$
$$5\bar{\alpha} + 0.44\delta_m = 0.757\,, \tag{55}$$

whose solution is:

$$\delta_m = -0.18\text{rad} = -10.5^{\text{o}} \tag{56}$$
$$\bar{\alpha} = 0.167\text{rad} = 9.6^{\text{o}} \tag{57}$$
$$\alpha = \bar{\alpha} + \alpha_0 = 0.167 - 0.035 = 0.132 \text{ rad} = 7.6^{\text{o}} \tag{58}$$

9. From the 1000 m steady-state, evaluate the slope created by deploying the landing gear.

Answer.

Deploying the landing gears only impacts C_x, so there won't be any change in velocity, only a change in descending slope (same as for the air brakes). Looking at $\boxed{\mathbf{P}}$, recalling that the thrust does not change, and with ΔD denoting the change in drag:

$$\Delta D = -mg\gamma \tag{59}$$
$$\Delta D = \frac{1}{2}\rho S V^2 \Delta C_{x_{0,LG}} = \frac{1}{2} 1.1 \times 260 \times 99.5^2 \times 0.0175 = 0.250 \times 10^5 N \tag{60}$$
$$\gamma = -\frac{\Delta D}{mg} = -\frac{0.25 \times 10^5}{10.8 \times 10^5} = -0.023 \text{ rad} = -1.32^{\text{o}} \tag{61}$$

The landing gears puts the aircraft on its final descent phase. It is not quite the regulatory ILS slope (3°), but we are halfway there.

10. Still from the 1000 m steady-state, let's evaluate the impact of deploying the slats and flaps (that is, assuming the landing gear are still retracted). To keep a realistic situation, assume that the pilot acts on δ_m to maintain the angle of attack constant during deployment (else, without acting on δ_m, you would end up with a negative C_z caused by the very negative C_{m_0}). Calculate both the new δ_m and the corresponding C_z, and compare it to the case with no slats and flaps. Comment on the value of deploying slats and flaps. Why would that be beneficial for the very last part of the flight, before landing?

Solution

We assume that the deployment of slats and flaps occurs at a constant angle of attack α, so that:

$$\bar{\alpha} = \alpha - \alpha_{0,SF} = 0.132 - (-0.262) = 0.394\text{rad} = 22.6^{\text{o}} \tag{62}$$

To maintain $C_m = 0$, we need to act on δ_m:

$$\delta_m = -\frac{C_{m_0} + \Delta C_{m_{0,SF}} + C_{m_\alpha}\bar{\alpha}}{C_{m_{\delta_m}}} \tag{63}$$

$$\delta_m = -\frac{-0.1 - 0.3 - 1(0.394)}{-1.46} = -\frac{0.794}{1.46} = -0.54\text{rad} = -30^{\text{o}} \tag{64}$$

This is a very large deflection for δ_m! This shows that to counter-act the strong C_{m_0} created by the slats and flaps, you need to have a significant δ_m margin. Note that on a number of aircrafts, the angle of the entire horizontal empennage can be adjusted (trim), so you absorb part of this δ_m by changing the trim.

Let's see the change in C_z that the slats and flaps have created:

$$C_z = C_{z_\alpha}\bar{\alpha} + C_{z_{\delta_m}}\delta_m = 5 \times 0.394 + 0.44(-0.54) = 1.97 - 0.24 = 1.73 \tag{65}$$

With the same angle of attack, C_z more than doubled, from 0.76 to 1.73! This increase will be leveraged to reduce velocity during the last phase.

4 Final approach, slats and flaps deployed

In this final approach stage, slats and flaps are deployed, along with landing gear, and the pilot positioned himself on the ILS beam, with a descent slope of 3°.

We want to evaluate that last equilibrium, and more specifically, the final approach velocity and the throttle required to maintain a constant slope. The approach velocity is still 30% above stall:

$$V_{ap} = 1.3V_{\mathrm{S}}\,, \tag{66}$$

but this time, stall is defined accounting for slats, flaps, and landing gears deployed. Calculate the new maximum lift coefficient $C_{z_{\max}}$, stall velocity V_{S}, approach velocity V_{ap}, angle of attack $\bar{\alpha}$, position of the elevator δ_m, lift and drag coefficients, and throttle δ_x. Comment on the throttle value you obtained.

Answer.

- The moment equation $\boxed{\mathbf{M}}$ is the one to use to calculate the δ_m needed to reach the max angle of attack, from which we can get $C_{z_{\max}}$.

$$C_m = C_{m_0} + \Delta C_{m_{0,SF}} + C_{m_\alpha}\bar{\alpha}_{\max} + C_{m_{\delta_m}}\delta_m \tag{67}$$

$$\delta_m = -\frac{C_{m_0} + \Delta C_{m_{0,SF}} + C_{m_\alpha}\bar{\alpha}_{\max}}{C_{m_{\delta_m}}} \tag{68}$$

$$\delta_m = -\frac{-0.1 - 0.3 - 1(0.576)}{-1.46} = -\frac{0.976}{1.46} = -0.668\mathrm{rad} = -38^{\mathrm{o}} \tag{69}$$

- The lift coefficient is

$$C_{z_{\max}} = C_{z_\alpha}\bar{\alpha}_{\max} + C_{z_{\delta_m}}\delta_m = 5 \times 0.576 + 0.44(-0.668) = 2.88 - 0.294 = 2.6 \tag{70}$$

  The max lift coefficient with slats and flaps deployed has been multiplied by two.

- The stall velocity comes from $\boxed{\mathbf{S}}$:

$$V_S^2 = \frac{mg}{\frac{1}{2}\rho S C_{z_{\max}}} = 2.9 \times 10^3 \tag{71}$$

$$V_S = 53.8m/s = 104.5kt \tag{72}$$

$$V_f = 1.3V_S = 70m/s = 136kt \tag{73}$$

- We can recover C_{z_f}, the corresponding angle of attack $\bar{\alpha}_f$, and the pitch control δ_{m_f} for this final phase:

$$\left(\frac{V_S}{V_f}\right)^2 = \frac{C_{z_f}}{C_{z_{\max}}} = \left(\frac{1}{1.3}\right)^2 = 0.592 \tag{74}$$

$$C_{z_f} = 2.6 \times 0.592 = 1.54 \tag{75}$$

$$\bar{\alpha}_f = 0.352\mathrm{rad} = 20.2^{\mathrm{o}} \tag{76}$$

$$\delta_{m_f} = -0.515\mathrm{rad} = -29.5^{\mathrm{o}} \tag{77}$$

- To evaluate the drag and δ_x, let's first evaluate the drag coefficient C_{x_f}:

$$C_{x_f} = C_{x_0} + \Delta C_{x_{0,f}} + kC_{z_f}^2 \tag{78}$$

$$\Delta C_{x_{0,f}} = \Delta C_{x_{0,LG}} + \Delta C_{x_{0,SF}} = 0.0175 + 0.06 = 0.0775 \tag{79}$$

$$C_{x_f} = 0.0175 + 0.0775 + 0.055 \times 1.53^2 = 0.224 \tag{80}$$

The drag and thrust, knowing that the slope is $\gamma = 3^o$ are:

$$D = \frac{1}{2}\rho S V^2 C_{x_f} = 0.5 \times 1.1 \times 260 \times 70^2 \times 0.224 = 1.57 \times 10^5 N \tag{81}$$

$$F = D + mg\gamma = 1.57 \times 10^5 - 0.556 \times 10^5 = 10^5 N \tag{82}$$

$$F = d\frac{\rho}{\rho_0} F_0 \delta_x = 2\frac{1.1}{1.225} 1.2 \times 10^5 \delta_x = 2.16 \times 10^5 \delta_x \tag{83}$$

This last result corresponds to a throttle control at 46% of max. This number is not negligible, which shows that high-lift systems are costly!